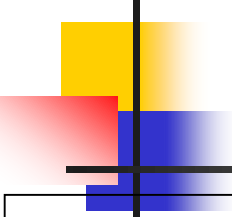


计量经济学

<< *Econometrics* >>

王琴英



复习：什么是相关关系（**Correlation**）

两类现象在发展变化的方向与大小方面存在一定的联系

——**MBA**智库百科

统计相关性

Pearson 线性相关

Spearman 等级相关

Kendall 协同相关

一般而言，相关关系指两变量之间线性相关的程度

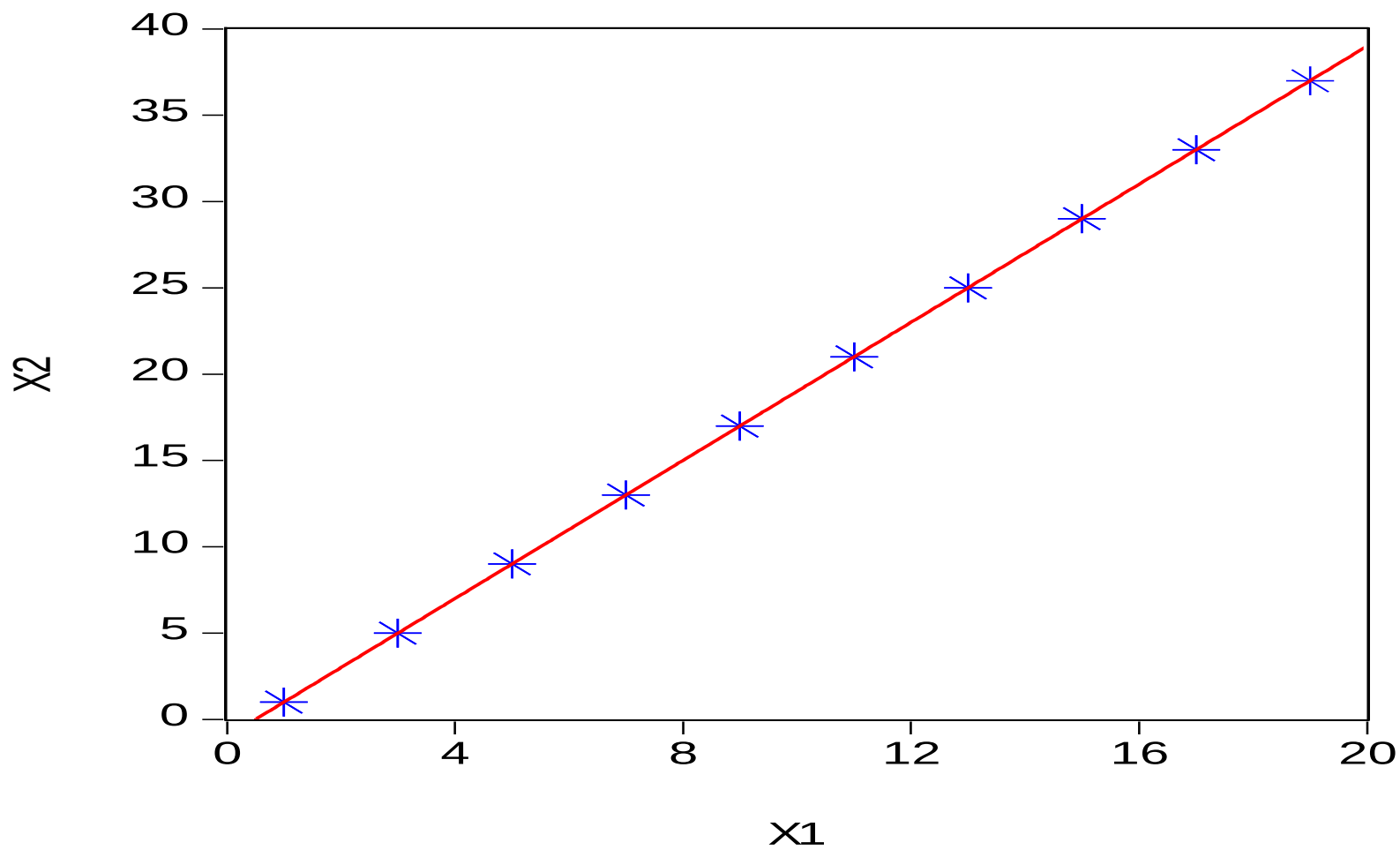


图1 X_1 与 X_2 之间完全正相关

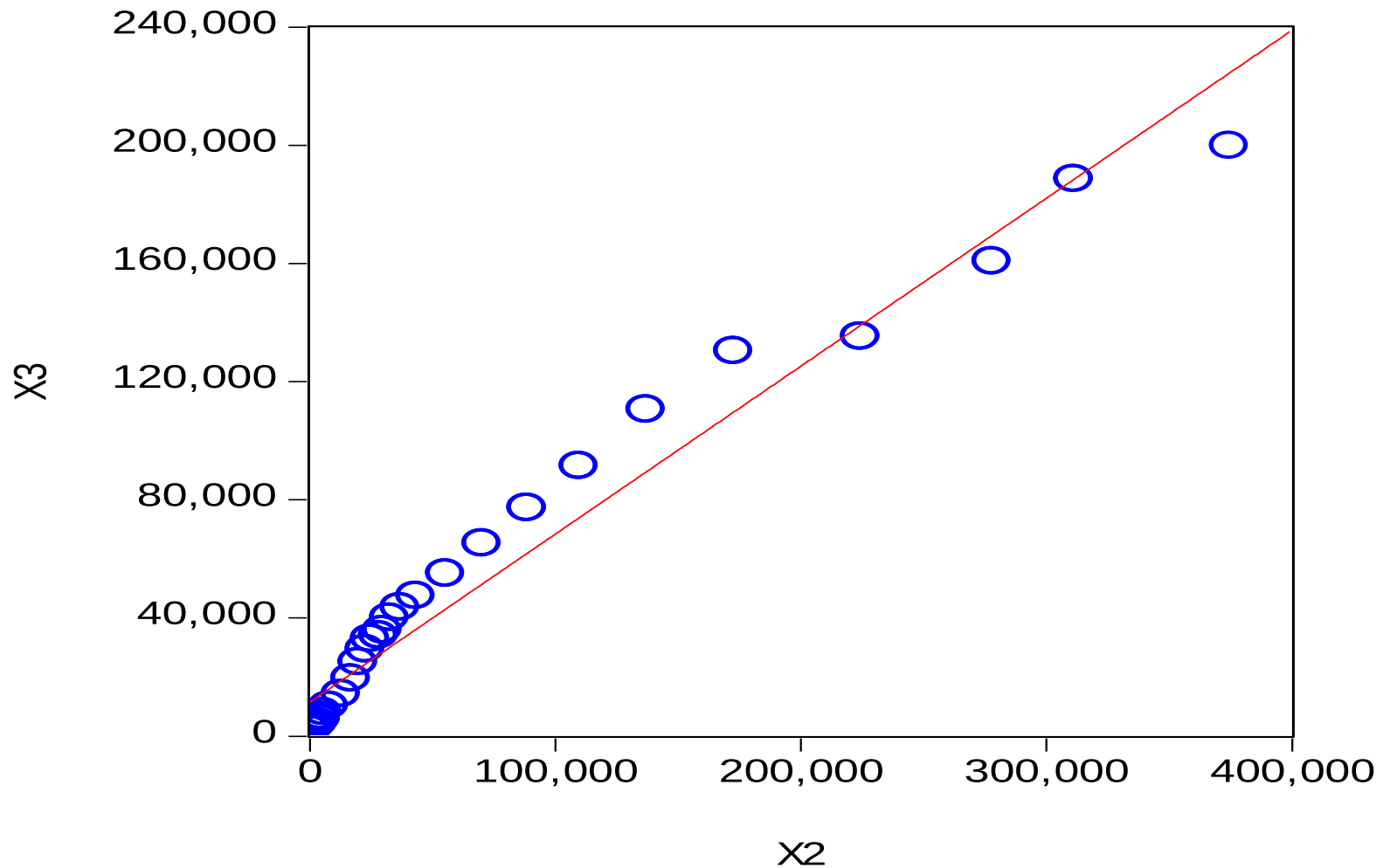


图2 X_2 与 X_3 之间近似正相关（高度正相关）

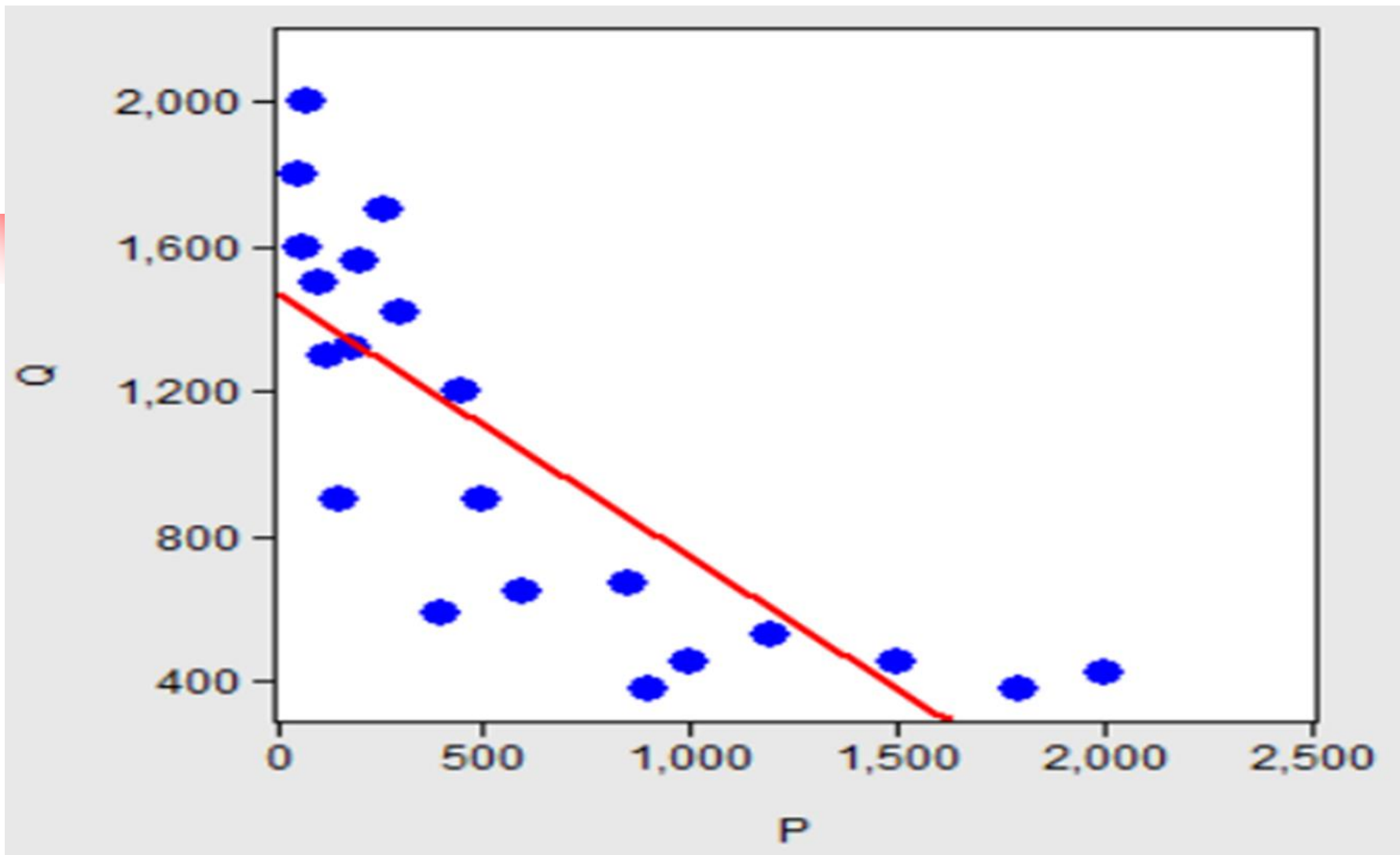


图3 销售额（Q）与价格水平（P）之间高度负相关

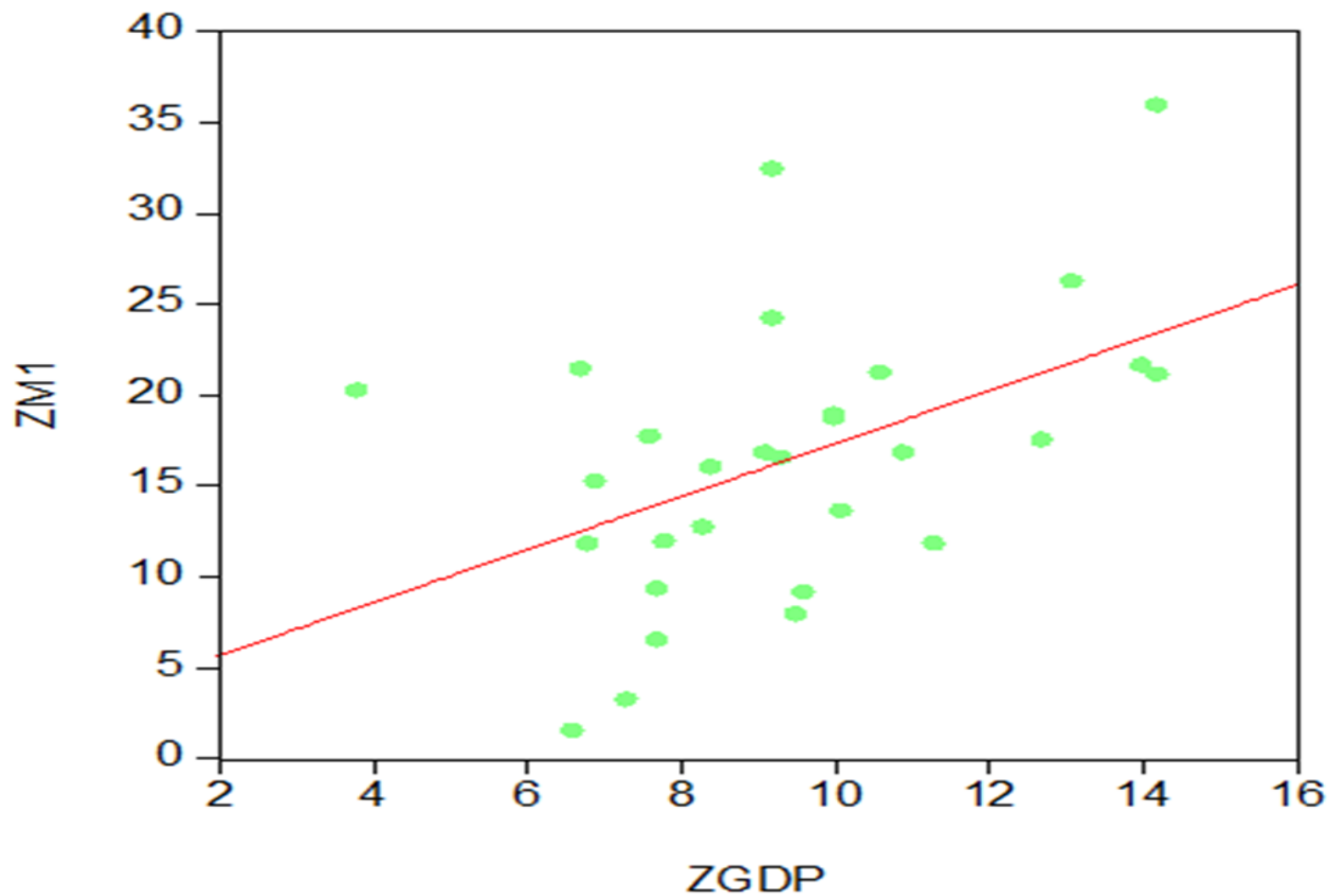
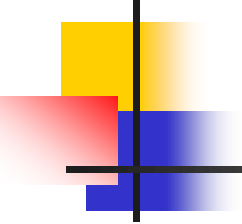


图4 ZM1与ZGDP之间几乎不相关



线性相关系数的取值特点:

- (1) 正相关: $r_{xy} > 0$; 完全正相关: $r_{xy} = 1$
- (2) 负相关: $r_{xy} < 0$; 完全负相关: $r_{xy} = -1$
- (3) 完全不相关: $r_{xy} = 0$
- (4) 高度相关: $|r_{xy}| \geq 0.8$ (经验值)
- (5) 显著相关: $|r_{xy}| > \gamma_{\alpha/2}(df)$ (临界值)
- (6) 不相关 (或相关性很弱) : $|r_{xy}|$ 很小

A. 7 检验相关系数 $\rho = 0$ 的临界值 (γ_α) 表

$$P(|\gamma| > \gamma_\alpha = \alpha$$

$f \backslash \alpha$	0.10	0.05	0.02	0.01	0.001	$\alpha \backslash f$
1	0.98769	0.999692	0.999507	0.999877	0.9999988	1
2	.90000	.9500	.98000	.99000	.99900	2
3	.8054	.8783	.93433	.95873	.99116	3
4	.7293	.8114	.88222	.91720	.97406	4
5	.6694	.7545	.8329	.8745	.95074	5
6	.6215	.7067	.7887	.8343	.92493	6
7	.5822	.6664	.7498	.7677	.8982	7
8	.5494	.6319	.7155	.7646	.8721	8
9	.5214	.6021	.6851	.7348	.8471	9
10	.4973	.5760	.6581	.7079	.8233	10
50	.2306	.2732	.3218	.3541	.4433	50
60	.2108	.2500	.2948	.3248	.4078	60
70	.1954	.2319	.2737	.3017	.3799	70
80	.1829	.2172	.2565	.2830	.3568	80
90	.1726	.2050	.2422	.2673	.3375	90
100	.1638	.1946	.2301	.2540	.3211	100

相关系数临界值表

第6章 多重共线性

6.1 多重共线性

一、多重共线性的含义

对于 **k** 元线性回归模型

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki} + u_i, \quad i = 1, 2, \cdots, n$$

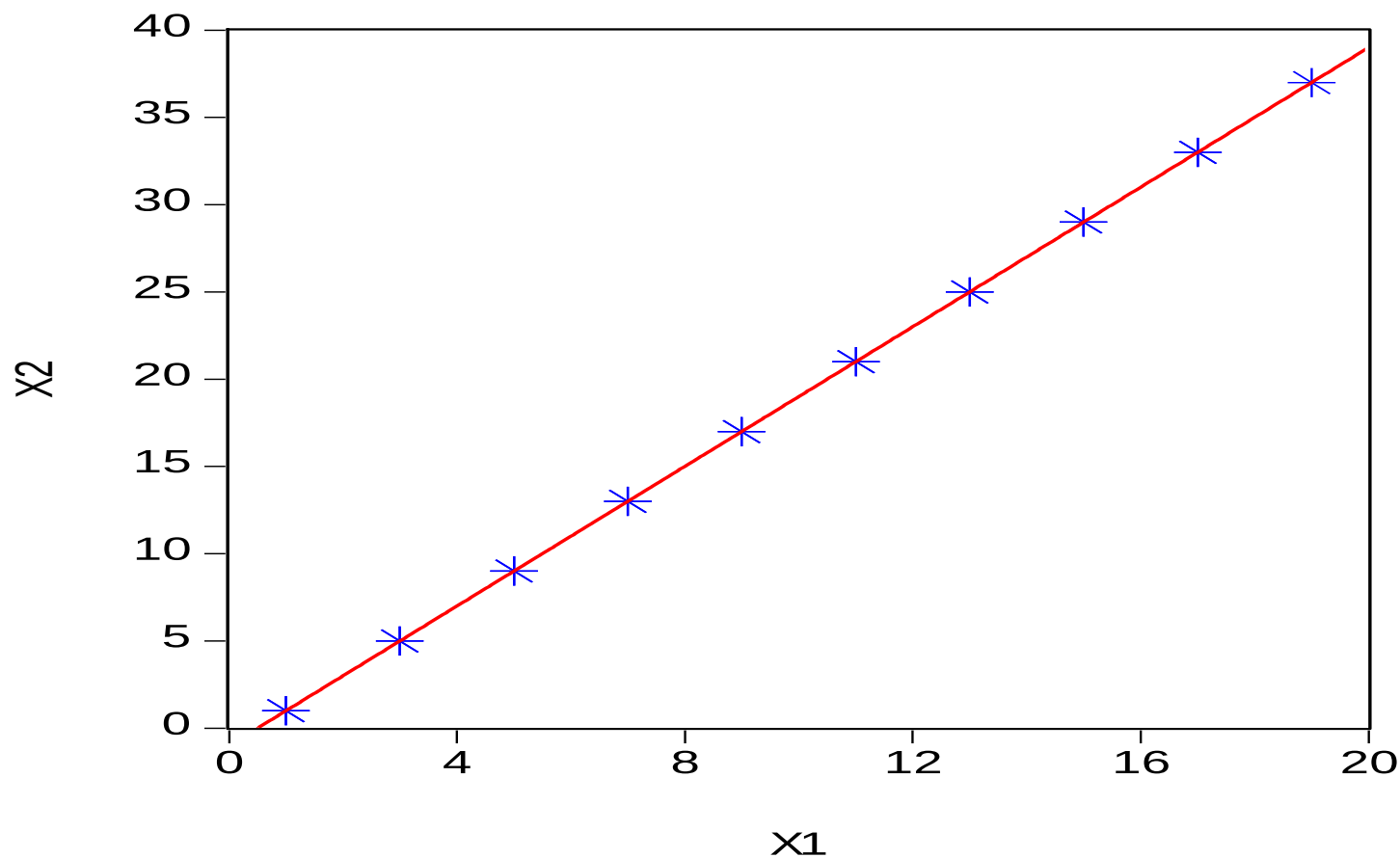
定义 两个或多个自变量之间出现相关，则称模型存在多重共线性

(1) 完全多重共线性: $X_1, X_2, \cdots, X_k$ 之间完全相关。

即存在不全为零的常数 λ_j ，使得

$$\lambda_1 X_1 + \lambda_2 X_2 + \cdots + \lambda_k X_k = a$$

自变量之间完全正相关



此直线关系如何用方程表示出来？

教材：表6-2	Y	X1	X2
	20	1	1
	18	3	5
	17	5	9
	15	7	13
	14	9	17
	12	11	21
	10	13	25
	11	15	29
	9	17	33
	7	19	37

$$X_{2i} = -1 + 2X_{1i} \quad , \quad i = 1, 2, \cdots, 10$$



(2) 不完全多重共线性

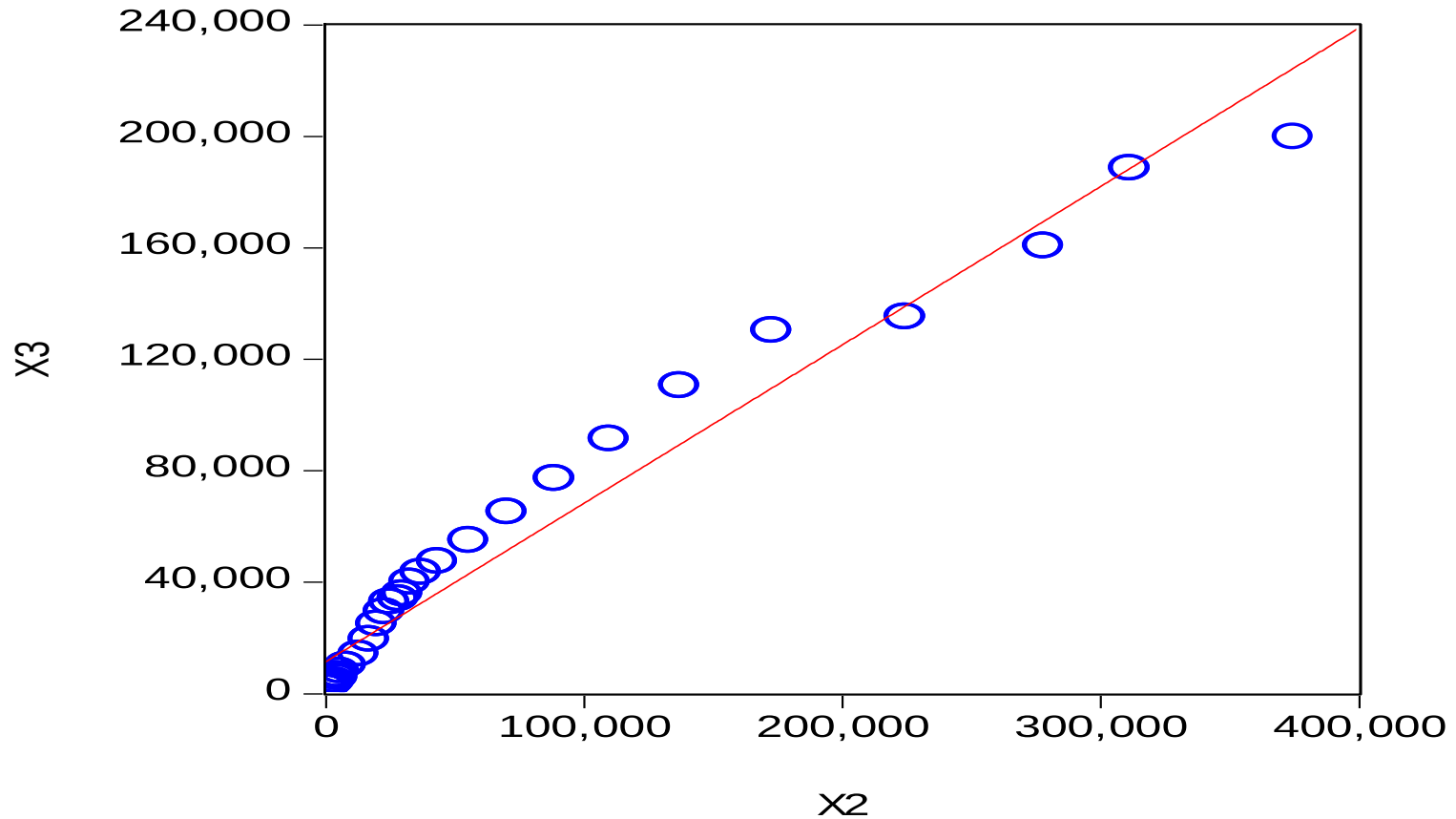
$X_1, X_2, \dots, X_k$ 之间存在近似的线性关系。

即存在不全为零的常数 λ_j ，使得

$$\lambda_1 X_1 + \lambda_2 X_2 + \dots + \lambda_k X_k + v = a$$

其中： V 为随机误差项。

自变量之间高度相关



近似直线关系，模型如何表示？回归直线如何表示？

二、产生多重共线性的原因

经济变量本身的特性，使自变量之间往往存在不同程度的线性相关。时间序列数据和截面数据都可能产生多重共线性

例如

1. 对于时间序列数据，研究消费与收入、财产的关系模型：

$$c_t = \beta_0 + \beta_1 I_t + \beta_2 w_t + u_t$$

一般有较高收入的家庭，其财产拥有量也较富裕。

财产与收入高度相关。

2. 运用截面数据，研究生产函数：

$$y_t = AK_t^{\beta_1} L_t^{\beta_2} e^{u_t}$$

由于资本投入和劳动投入都与企业的生产规模有关，统计数据往往表明资本投入与劳动投入之间存在显著相关。

6.2 多重共线性引起的后果

一、完全共线性的情形

对于二元线性模型: $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i \dots (1)$

设 $X_{2i} = a + bX_{1i}$ (a, b 为已知的常数)

$$\Rightarrow \bar{x}_2 = a + b\bar{x}_1, \quad \dot{x}_2 = b\dot{x}_1, \quad \sum \dot{x}_{2i}^2 = b^2 \sum \dot{x}_{1i}^2$$

$$\sum \dot{x}_{1i} \dot{x}_{2i} = b \sum \dot{x}_{1i}^2, \quad \sum \dot{x}_{2i} \dot{y}_i = b \sum \dot{x}_{1i} \dot{y}_i$$

可证: 在此情形下, 未知参数的最小二乘估计值不存在

事实上

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum \dot{x}_{2i}^2 \sum \dot{x}_{1i} \dot{y}_i - \sum \dot{x}_{1i} \dot{x}_{2i} \sum \dot{x}_{2i} \dot{y}_i}{\sum \dot{x}_{1i}^2 \sum \dot{x}_{2i}^2 - (\sum \dot{x}_{1i} \dot{x}_{2i})^2} \\&= \frac{b^2 \sum \dot{x}_{1i}^2 \sum \dot{x}_{1i} \dot{y}_i - (b \sum \dot{x}_{1i}^2) (b \sum \dot{x}_{1i} \dot{y}_i)}{\sum \dot{x}_{1i}^2 (b^2 \sum \dot{x}_{1i}^2) - (b \sum \dot{x}_{1i}^2)^2} \\&= \frac{0}{0}\end{aligned}$$

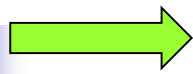
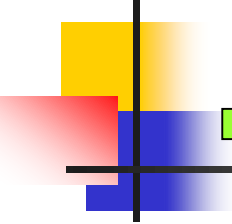
用类似方法可证 $\hat{\beta}_2 = \frac{0}{0}$ 。可见 $\hat{\beta}_1$ 和 $\hat{\beta}_2$ 完全不确定

由于 $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} - \hat{\beta}_2 \bar{x}_2$ ， $\hat{\beta}_0$ 同样完全不确定。

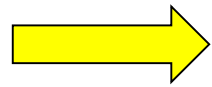
因此，在完全多重共线的情形下，最小二乘估计量不存在

比如： 对于 $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$

设 $x_{2i} = 0.6 + 3 x_{1i}$


$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_{1i} + \beta_2 (0.6 + 3 x_{1i}) + u_i \\ &= (\beta_0 + 0.6 \beta_2) + (\beta_1 + 3 \beta_2) x_{1i} + u_i \end{aligned}$$

$$\diamond A_0 = (\beta_0 + 0.6 \beta_2), A_1 = (\beta_1 + 3 \beta_2)$$


$$y_i = A_0 + A_1 x_{1i} + u_i$$

$$OLS: \hat{y}_i = \hat{A}_0 + \hat{A}_1 x_{1i} = 1.21 + 0.58 x_{1i}$$

$$\begin{cases} \hat{\beta}_0 + 0.6 \hat{\beta}_2 = 1.21 \\ \hat{\beta}_1 + 3 \hat{\beta}_2 = 0.58 \end{cases} \Rightarrow \hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2 ??$$

由**2**个方程求解**3**个未知数，是无解还是有无穷多解？

二、不完全多重共线性的情形

设 X_2 与 X_1 高度相关，不妨设模型为：

$$X_{2i} = a + bx_{1i} + v_i \quad \dots (2)$$

对模型（2）用最小二乘法估计未知参数，得到估计方程：

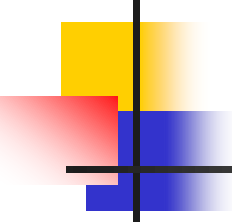
$$\hat{X}_{2i} = \hat{a} + \hat{b}x_{1i}$$

记 $\varepsilon_i^* = \hat{v}_i = X_{2i} - \hat{X}_{2i}$ 为模型（2）的残差；

最小二乘法要求此残差满足如下关系：

$$\left\{ \begin{array}{l} \sum \varepsilon_i^* = 0 \\ \sum \varepsilon_i^* x_{1i} = 0 \end{array} \right.$$

现将与 x_{2i} 有关的式子，均用 x_{1i} 来表示：


$$(1) \quad x_{2i} = \hat{a} + \hat{b}x_{1i} + \varepsilon_i^*$$

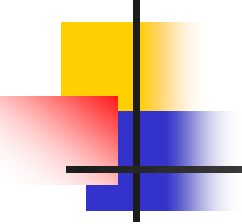
$$(2) \quad \bar{x}_2 = \hat{a} + \hat{b}\bar{x}_1$$

$$(3) \quad \dot{x}_{2i} = \hat{b}\dot{x}_{1i} + \varepsilon_i^*$$

$$(4) \quad \sum \dot{x}_{2i}^2 = \hat{b}^2 \sum \dot{x}_{1i}^2 + \sum \varepsilon_i^{*2}$$

$$(5) \quad \sum \dot{x}_{1i} \dot{x}_{2i} = \hat{b} \sum \dot{x}_{1i}^2$$

$$(6) \quad \sum \dot{x}_{2i} \dot{y}_i = \hat{b} \sum \dot{x}_{1i} \dot{y}_i + \sum \dot{y}_i \varepsilon_i^*$$



模型: $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$
OLS: $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i}$

$$\hat{\beta}_1 = \frac{\sum \dot{x}_{1i} \dot{y}_i}{\sum \dot{x}_{1i}^2} - \hat{b} \frac{\sum \dot{y}_i \varepsilon_i^*}{\sum \varepsilon_i^{*2}}$$


$$\text{Var}(\hat{\beta}_1) = \frac{\sigma_u^2 / \sum \dot{x}_{1i}^2}{(1 - r_{12}^2)} = \frac{\sigma_u^2 (\hat{b}^2 + \frac{\sum \varepsilon_i^{*2}}{\sum \dot{x}_{1i}^2})}{\sum \varepsilon_i^{*2}}$$

模型: $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$

OLS: $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i}$

不完全多重共线性产生的后果

1. **OLS**估计值及估计量的方差均跟自变量之间的回归残差 ε_i^* 有关, 因而, 它们对样本很敏感, 缺乏稳定性。
2. 估计量的方差较大。随着自变量的相关性的增强, 方差也随之增大。因此, **OLS**估计量不满足最佳性。
且当 $|r_{12}| \rightarrow 1$ 时, $Var(\hat{\beta}_1) \rightarrow \infty$
3. 参数的显著性检验失效; 预测失效。

6.3 多重共线性的检验

一、不显著系数法

1. 理论上，自变量对因变量有重要影响，但其系数的显著性检验却不显著。
2. 在原模型中，增加一个新的自变量后，参数的最小二乘估计量的方差明显增大。而方差（或标准方差）的增大可能导致显著性检验不显著。
3. 模型回归的拟合优度很大，但模型中全部或部分参数的显著性检验却不显著。

若出现上述三种情形之一，可以经验地认为自变量之间存在显著相关，模型存在多重共线性。



例1

随机调查某地区**14**户居民家庭每周家庭消费支出（**Y**）、每周家庭收入（**X1**）和 家庭财产拥有量（**X2**）的截面数据（单位：美元）如下表所示。试构建恰当的模型关系，以估计家庭消费支出对于家庭收入和财产拥有量的关系。



序号	y_i	x_{1i}	x_{2i}	序号	y_i	x_{1i}	x_{2i}
1	70	80	810	8	140	220	2201
2	65	100	1009	9	155	240	2435
3	90	120	1273	10	150	260	2686
4	95	140	1425	11	162	280	2921
5	110	160	1633	12	178	300	3146
6	115	180	1876	13	180	320	3500
7	120	200	2052	14	189	340	3890

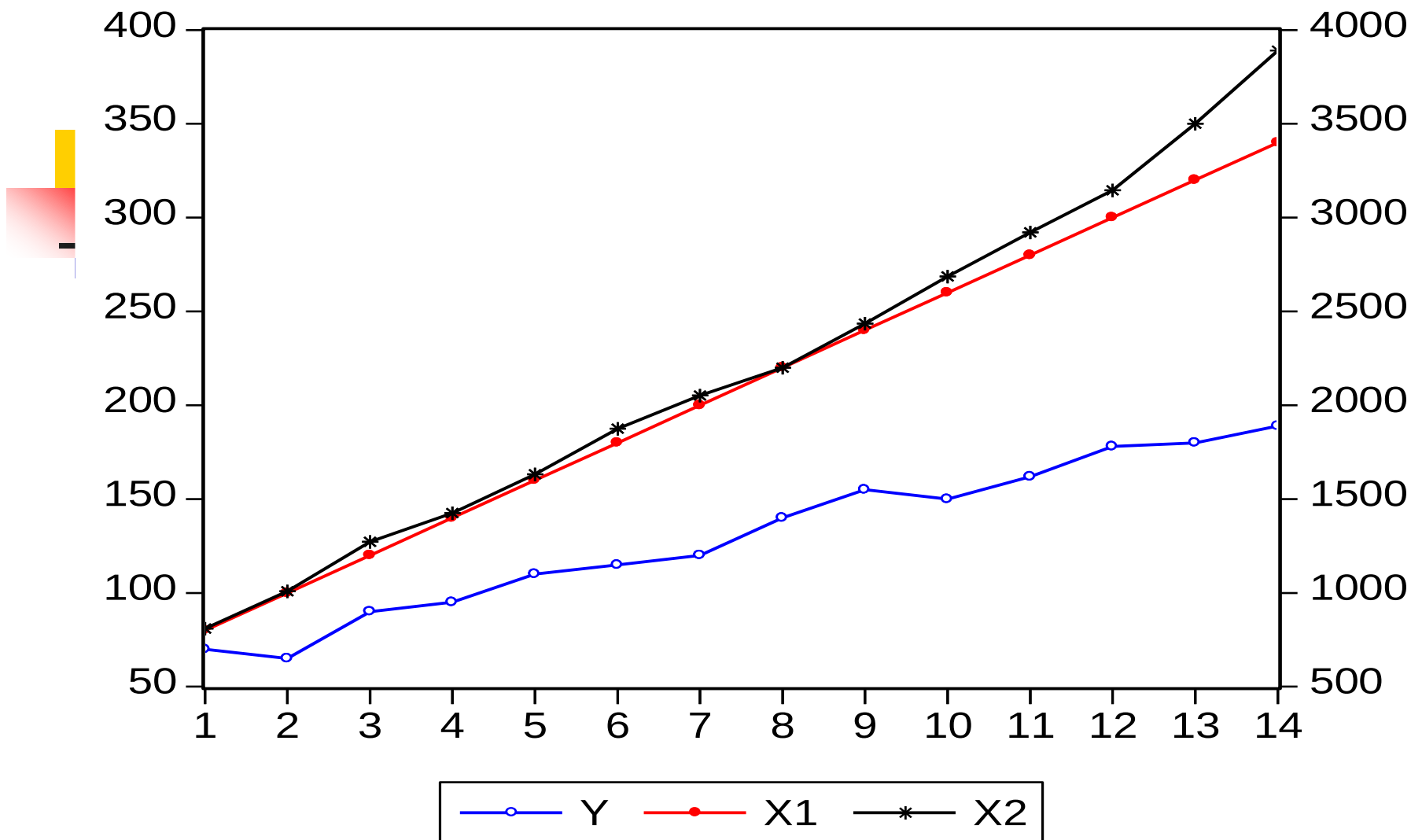


图1 折线图

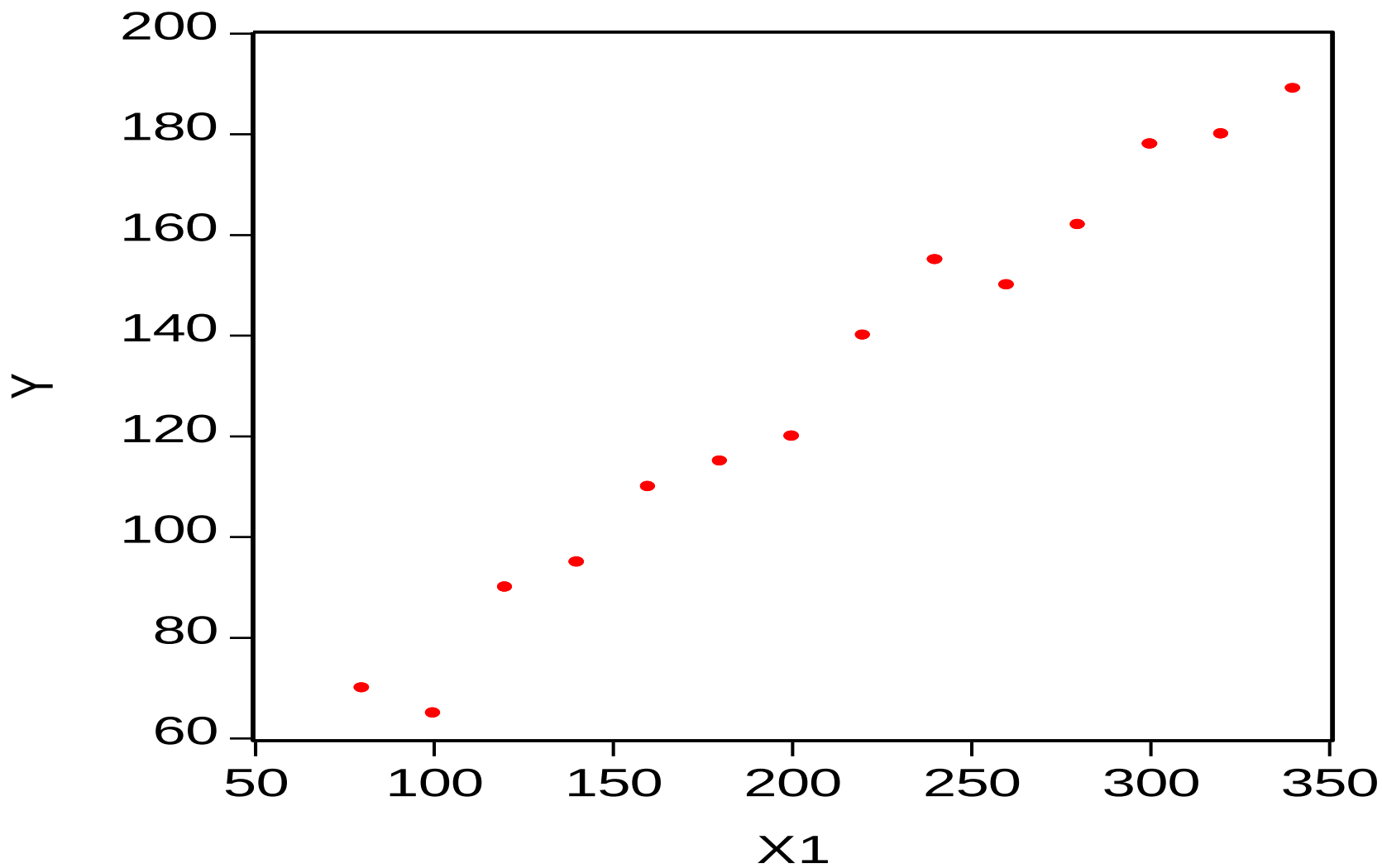


图2 (X1,Y) 散点图

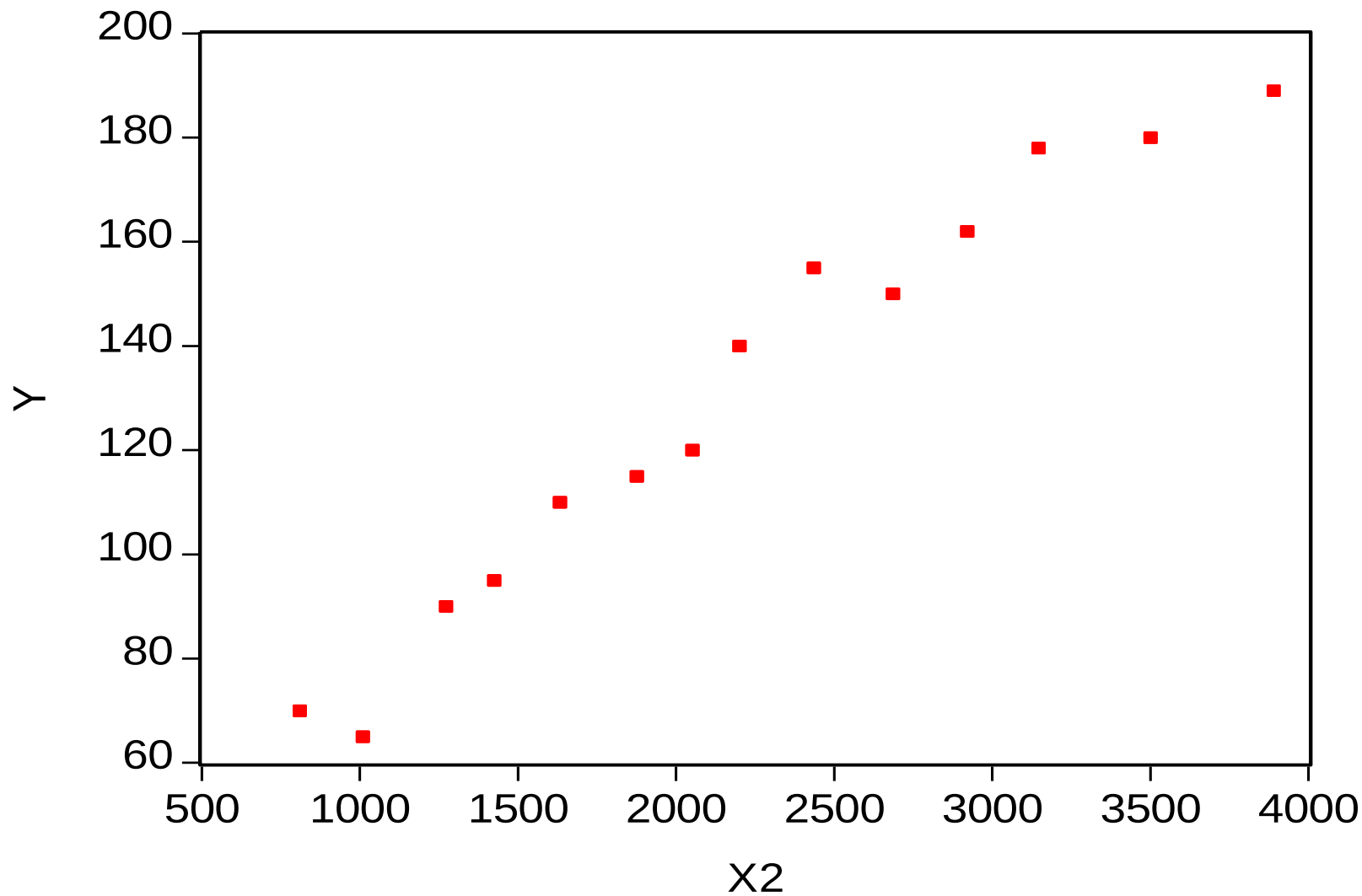


图3 (X_2, Y) 散点图

Dependent Variable: Y
Method: Least Squares

一元回归输出结果（一）

Sample: 1 14
Included observations: 14

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	28.09011	4.349661	6.458000	0.0000
X1	0.484945	0.019337	25.07913	0.0000
R-squared	0.981278	Mean dependent var	129.9286	
Adjusted R-squared	0.979718	S.D. dependent var	40.95864	
S.E. of regression	5.833119	Akaike info criterion	6.496544	
Sum squared resid	408.3033	Schwarz criterion	6.587838	
Log likelihood	-43.47581	F-statistic	628.9626	
Durbin-Watson stat	2.448068	Prob(F-statistic)	0.000000	

Dependent Variable: Y
Method: Least Squares

一元回归输出结果（二）

Sample: 1 14
Included observations: 14

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	35.81827	5.816432	6.158118	0.0000
X2	0.042698	0.002440	17.49583	0.0000
R-squared	0.962276	Mean dependent var	129.9286	
Adjusted R-squared	0.959133	S.D. dependent var	40.95864	
S.E. of regression	8.280044	Akaike info criterion	7.197137	
Sum squared resid	822.7096	Schwarz criterion	7.288431	
Log likelihood	-48.37996	F-statistic	306.1039	
Durbin-Watson stat	1.432963	Prob(F-statistic)	0.000000	

Dependent Variable: Y
Method: Least Squares

二元回归输出结果

Sample: 1 14

Included observations: 14

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	24.77148	4.910949	5.044133	0.0004
X1	0.738393	0.192696	3.831911	0.0028
X2	-0.022642	0.017133	-1.321555	0.2131
R-squared	0.983843	Mean dependent var	129.9286	
Adjusted R-squared	0.980906	S.D. dependent var	40.95864	
S.E. of regression	5.659731	Akaike info criterion	6.492039	
Sum squared resid	352.3582	Schwarz criterion	6.628980	
Log likelihood	-42.44428	F-statistic	334.9181	
Durbin-Watson stat	2.819002	Prob(F-statistic)	0.000000	

自变量**X2**的回归系数为何是负数？**T**检验显著吗

一元与二元线性回归结果比较



(1) $\hat{y}_i = 28.0901 + 0.4849x_{1i}$, $R^2 = 0.9813$

$S_{\hat{\beta}_i} :$ (4.3497) (0.0193)

$t :$ (6.4580) (25.0791)

(2) $\hat{y}_i = 24.7715 + 0.7384x_{1i} - 0.0226x_{2i}$

$S_{\hat{\beta}_j} :$ (4.9109) (0.1927) (0.0171)

$t :$ (5.0441) (3.8319) (-1.3216)

$R^2 = 0.9838$, $\bar{R}^2 = 0.9809$, $F = 334.9181$

DW= 2.819 , $r_{12} = 0.9952$

(3) $\hat{y}_i = 35.8183 + 0.0427x_{2i}$, $R^2 = 0.9623$

$S_{\hat{\beta}_1} :$ (5.8164) (0.0024)

$t :$ (6.1581) (17.4958)

(2) $\hat{y}_i = 24.7715 + 0.7384x_{1i} - 0.0226x_{2i}$

$S_{\hat{\beta}_j} :$ (4.9109) (0.1927) (0.0171)

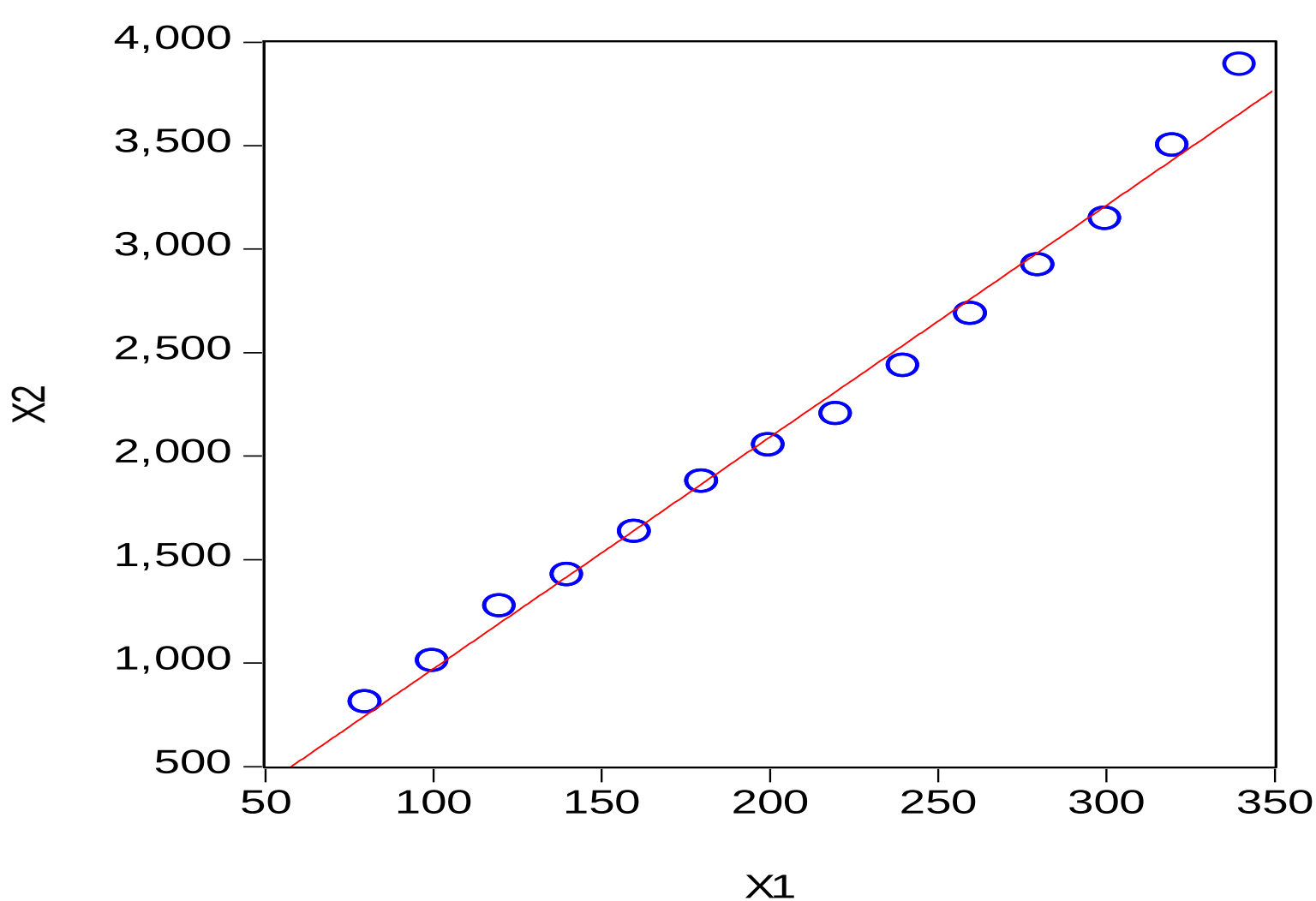
$t :$ (5.0441) (3.8319) (-1.3216)

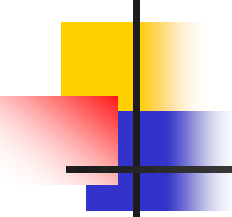
$R^2 = 0.9838$, $\bar{R}^2 = 0.9809$, $F = 334.9181$

DW= 2.819 , $r_{12} = 0.9952$

例1:

自变量之间几乎完全相关





例6-1

例 6-1 为探讨中国能源消费总量 y （单位：万吨标准煤）的主要影响因素，考虑生活能源消费量 x_1 （单位：万吨标准煤）、全社会固定资产投资 x_2 （单位：亿元）以及工业生产总值 x_3 （单位：亿元）三个因素对其影响的弹性大小。已知 1980~2012 年各变量的统计数据如表 6-1 所示，其中，生活能源消费量=人均年生活消费能源量×人口数量。（数据来源：国家统计局《统计年鉴》（2000、2013））。

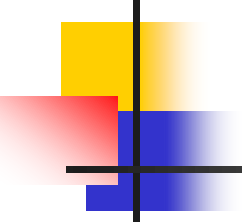
- 
-
- y —— 能源消费总量（单位：万吨标准煤）^{*}
- x_1 —— 生活能源消费量（单位：万吨标准煤）
- x_2 —— 全社会固定资产投资（单位：亿元）^{*}
- x_3 —— 工业生产总产值（单位：亿元）^{*}

表 6-1 y 、 x_1 、 x_2 与 x_3 的统计数据

年份	y	x_1	x_2	x_3
1980	60 275.0	8 120.46	910.90	1 996.50
1981	63 248.24	8 975.46	961.0	2 048.40
1982	66 368.15	9 939.73	1 230.40	2 162.30
1983	69 641.95	10 982.18	1 430.10	2 375.60
1984	73 077.25	11 839.52	1 832.90	2 789.0
1985	76 682.0	13 412.65	2 543.20	3 448.70
1986	80 850.0	13 688.43	3 120.60	3 967.0
1987	86 632.0	14 441.45	3 791.70	4 585.80
1988	92 997.0	15 655.69	4 753.80	5 777.20
1989	96 934.0	15 699.87	4 410.40	6 484.0
1990	98 703.0	15 799.0	4 517.0	6 858.0
1991	103 783.0	16 096.54	5 594.50	8 087.10
1992	109 170.0	15 726.46	8 080.10	10 284.50
1993	115 993.0	15 820.84	13 072.30	14 187.97
1994	122 737.0	15 499.19	17 042.10	19 480.71
1995	131 176.0	15 745.0	20 019.30	24 950.61
1996	135 192.0	14 748.63	22 974.0	29 447.61
1997	135 909.0	14 750.95	24 941.10	32 921.39
1998	136 184.0	14 847.35	28 406.20	34 018.43
1999	140 569.0	15 320.53	29 854.70	35 861.48
2000	145 531.0	15 614.0	32 917.73	40 033.59
2001	150 406.0	16 239.31	37 213.49	43 580.62
2002	159 431.0	17 217.82	43 499.91	47 431.31
2003	183 792.0	19 824.03	55 566.61	54 945.53
2004	213 456.0	22 835.25	70 477.40	65 210.03
2005	235 997.0	25 305.0	88 773.60	77 230.78
2006	258 676.0	27 838.44	109 998.20	91 310.94
2007	280 508.0	30 893.51	137 323.90	110 534.90
2008	291 448.0	31 979.35	172 828.40	130 260.20
2009	306 647.0	33 843.0	224 598.80	135 239.90
2010	324 939.0	34 558.0	278 121.90	160 722.20
2011	348 002.0	37 410.0	311 485.10	188 470.20
2012	361 732.0	39 332.0	374 694.70	199 670.70

资料来源：国家统计局《统计年鉴》(2000、2013)。

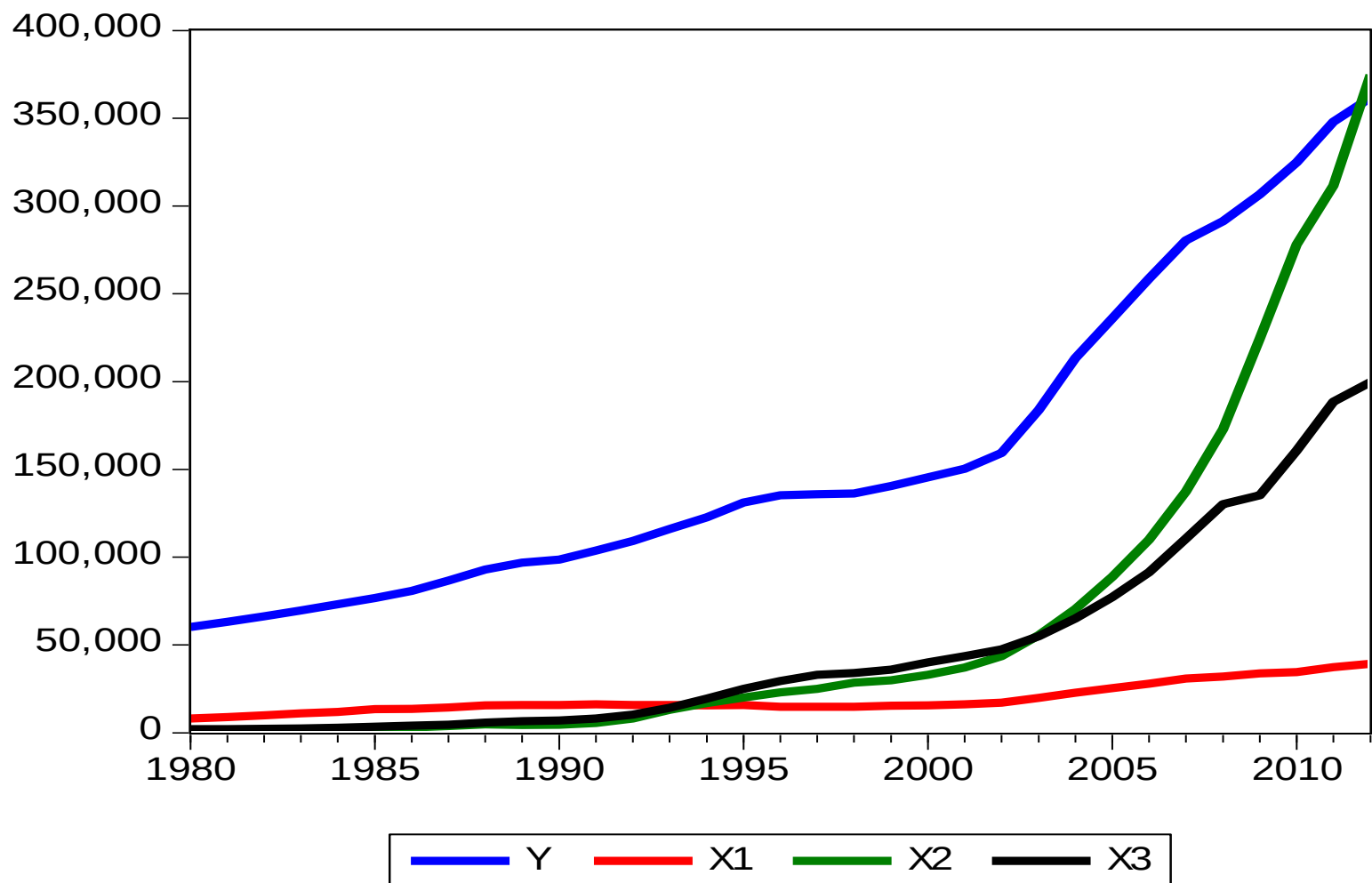
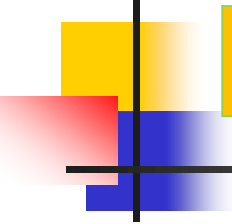


图4 折线图



假设模型关系：

双对数模型

$$\ln y_i = \beta_0 + \beta_1 \ln x_{1i} + \beta_2 \ln x_{2i} + \beta_3 \ln x_{3i} + u_i$$

为何设定成双对数模型？这类模型有什么特点？

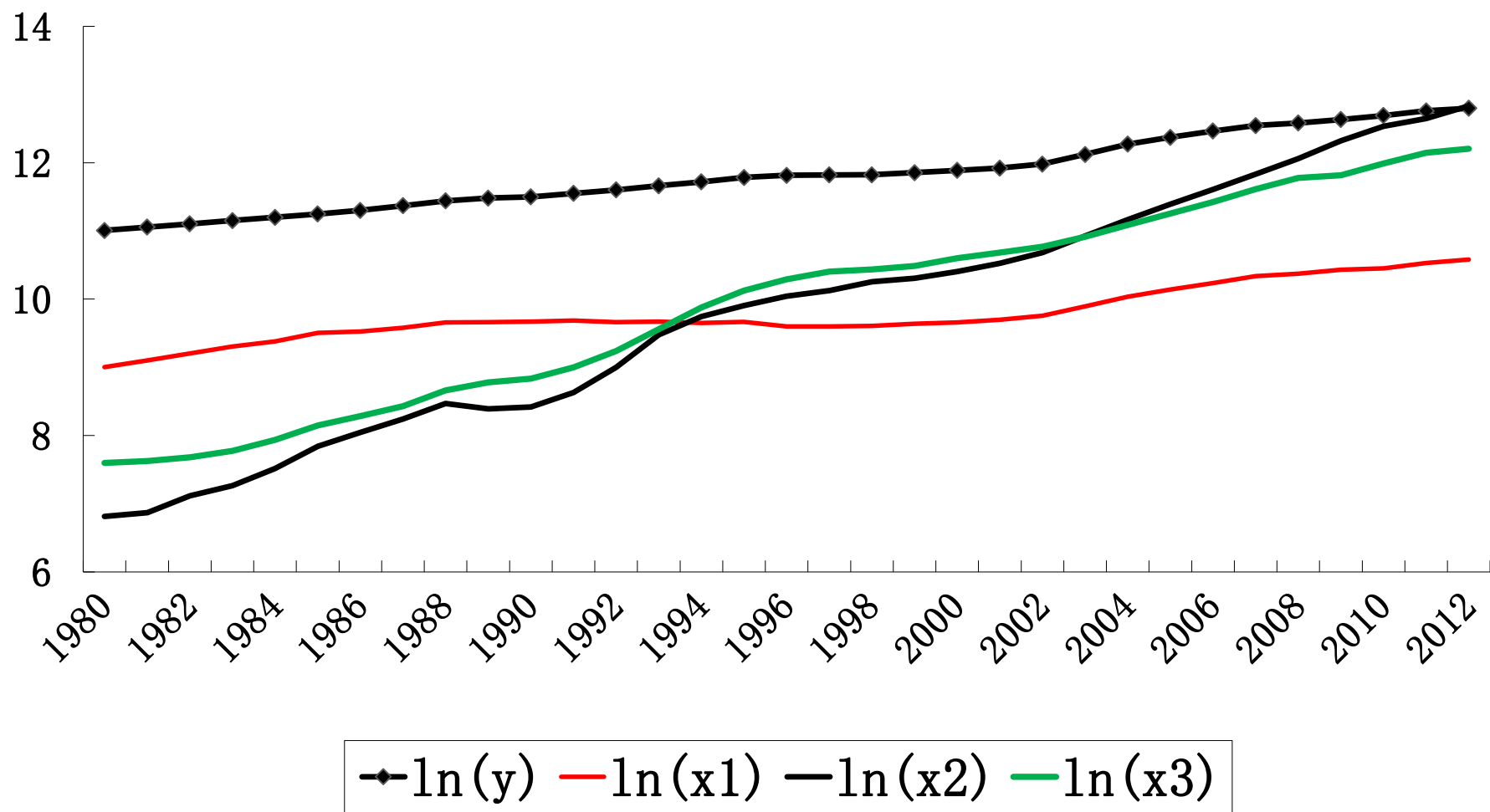


图5 对数变量的折线图

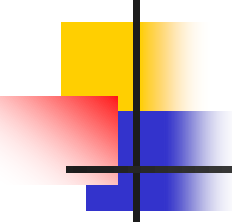
Dependent Variable: LOG(Y)
Method: Least Squares

三元回归结果

Sample: 1980 2012
Included observations: 33

	Coefficient	Std. Error	t-Statistic	Prob.
C	3.752801	0.797201	4.707470	0.0001
LOG(X1)	0.605661	0.082836	7.311576	0.0000
LOG(X2)	-0.057025	0.095930	-0.594447	0.5568
LOG(X3)	0.274255	0.099458	2.757497	0.0100
R-squared	0.993800	Mean dependent var	11.83346	
Adjusted R-squared	0.993159	S.D. dependent var	0.539974	
S.E. of regression	0.044662	Akaike info criterion	-3.266158	
Sum squared resid	0.057847	Schwarz criterion	-3.084763	
Log likelihood	57.89160	Hannan-Quinn criter.	-3.205124	
F-statistic	1549.494	Durbin-Watson stat	0.226723	
Prob(F-statistic)	0.000000			

自变量回归系数是负数，合理吗？其 **t** 检验显著吗



回归结果的综合表示

$$\ln \hat{y}_i = 3.7528 + 0.6057 \ln x_{1i} - 0.0570 \ln x_{2i} + 0.2743 \ln x_{3i}$$

$$t: \quad (4.7075) \quad (7.3116) \quad (-0.5944) \quad (2.7575)$$

$$P: \quad (0.0001) \quad (0.0000) \quad (0.5568) \quad (0.0100)$$

$$R^2 = 0.994, \quad \bar{R}^2 = 0.993, \quad F = 1549.492$$



二、相关系数法

X_1 与 X_2 之间的相关系数为:

$$r_{12} = \frac{\sum \dot{x}_{1i} \dot{x}_{2i}}{\sqrt{(\sum \dot{x}_{1i}^2)(\sum \dot{x}_{2i}^2)}}$$

A.7 检验相关系数 $\rho=0$ 的临界值 (γ_α) 表

$$P(|\gamma| > \gamma_\alpha = \alpha)$$

$f \backslash \alpha$	0.10	0.05	0.02	0.01	0.001	$\alpha \backslash f$
1	0.98769	0.999692	0.999507	0.999877	0.9999988	1
2	.90000	.9500	.98000	.99000	.99900	2
3	.8054	.8783	.93433	.95873	.99116	3
4	.7293	.8114	.88222	.91720	.97406	4
5	.6694	.7545	.8329	.8745	.95074	5
6	.6215	.7067	.7887	.8343	.92493	6
7	.5822	.6664	.7498	.7677	.8982	7
8	.5494	.6319	.7155	.7646	.8721	8
9	.5214	.6021	.6851	.7348	.8471	9
10	.4973	.5760	.6581	.7079	.8233	10
11	.4762	.5529	.6339	.6835	.8010	11
12	.4575	.5324	.6120	.6614	.7800	12
13	.4409	.5139	.5923	.6411	.7103	13
14	.4259	.4973	.5742	.6226	.7420	14
15	.4124	.4821	.5577	.6055	.7246	15

临界值表

例6-1的变量间相关系数

Correlation				
	Y	X1	X2	X3
Y	1.000000	0.981962	0.938313	0.982978
X1	0.981962	1.000000	0.945037	0.968952
X2	0.938313	0.945037	1.000000	0.980805
X3	0.982978	0.968952	0.980805	1.000000

Correlation				
	LOG(Y)	LOG(X1)	LOG(X2)	LOG(X3)
LOG(Y)	1.000000	0.961107	0.988552	0.978636
LOG(X1)	0.961107	1.000000	0.925367	0.896000
LOG(X2)	0.988552	0.925367	1.000000	0.996246
LOG(X3)	0.978636	0.896000	0.996246	1.000000

三、判定系数（即拟合优度）检验法

对于模型 $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki} + u_i$

为找出引起共线性的自变量，将模型中每一个自变量对其余自变量作回归，以检验自变量之间的这种线性关系是否整体显著？

作检验统计量：

$$F_j = \frac{ESS / (k-1)}{RSS / (n-k)} = \frac{R_j^2 / (k-1)}{(1-R_j^2) / (n-k)} \quad j=1, 2, \dots, k$$

用此检验第 j 个自变量与其余 $k-1$ 个自变量之间是否存在显著的共线性。

R_j^2 为第 j 个自变量对其余自变量进行回归的拟合优度



自变量之间的回归模型整体显著性检验

对于模型: $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki} + u_i$

设 $x_{2i} = a_0 + a_1 x_{1i} + a_3 x_{3i} + \cdots + a_k x_{ki} + v_i$

$H_0: a_1 = a_3 = \cdots = a_k = 0$, $H_1: a_1, a_3, \cdots, a_k$ 至少有一个不为零

问: 如何完成此检验?

检验方法与过程



设 $x_{2i} = a_0 + a_1 x_{1i} + a_3 x_{3i} + \cdots + a_k x_{ki} + v_i$

(1) $OLS: \hat{x}_{2i} = \hat{a}_0 + \hat{a}_1 x_{1i} + \hat{a}_3 x_{3i} + \cdots + \hat{a}_k x_{ki}$; 拟合优度 R^2

(2) $H_0: a_1 = a_3 = \cdots = a_k = 0$, $H_1: a_1, a_3, \cdots, a_k$ 至少有一个不为零

(3) 作
$$F_2 = \frac{ESS / (k-1)}{RSS / [(n-(k-1)) - 1]} = \frac{R^2 / (k-1)}{(1-R^2) / (n-k)}$$

(4) H_0 的拒绝域为: $F_2 > F_\alpha((k-1), (n-k))$

或: $p\text{值} = P_{H_0}(F_2 > F_\alpha((k-1), (n-k))) \leq \alpha$

(5) 推断:

例6-1（续）找出引起共线性的自变量

Dependent Variable: LOG(X1)

Method: Least Squares

Sample: 1980 2012

Included observations: 33

第1个自变量与其余自变量的回归

	Coefficient	Std. Error	t-Statistic	Prob.
C	9.508631	0.271086	35.07601	0.0000
LOG(X2)	0.986015	0.110892	8.891704	0.0000
LOG(X3)	-0.947336	0.134678	-7.034072	0.0000
R-squared	0.945760	Mean dependent var	9.771714	
Adjusted R-squared	0.942144	S.D. dependent var	0.409251	
S.E. of regression	0.098438	Akaike info criterion	-1.712272	
Sum squared resid	0.290701	Schwarz criterion	-1.576226	
Log likelihood	31.25250	Hannan-Quinn criter.	-1.666497	
F-statistic	261.5505	Durbin-Watson stat	0.678989	
Prob(F-statistic)	0.000000			

Dependent Variable: LOG(X2)
Method: Least Squares

第2个自变量与其余自变量的回归

Sample: 1980 2012
Included observations: 33

	Coefficient	Std. Error	t-Statistic	Prob.
C	-7.597499	0.614765	-12.35838	0.0000
LOG(X1)	0.735210	0.082685	8.891704	0.0000
LOG(X3)	1.029313	0.022673	45.39913	0.0000
R-squared	0.997938	Mean dependent var	9.799517	
Adjusted R-squared	0.997801	S.D. dependent var	1.812657	
S.E. of regression	0.085002	Akaike info criterion	-2.005787	
Sum squared resid	0.216758	Schwarz criterion	-1.869741	
Log likelihood	36.09548	Hannan-Quinn criter.	-1.960011	
F-statistic	7261.081	Durbin-Watson stat	0.878728	
Prob(F-statistic)	0.000000			

Dependent Variable: LOG(X3)
Method: Least Squares

第3个自变量与其余自变量的回归

Sample: 1980 2012
Included observations: 33

	Coefficient	Std. Error	t-Statistic	Prob.
C	6.959489	0.726016	9.585865	0.0000
LOG(X1)	-0.657145	0.093423	-7.034072	0.0000
LOG(X2)	0.957583	0.021093	45.39913	0.0000
R-squared	0.997171	Mean dependent var	9.921906	
Adjusted R-squared	0.996982	S.D. dependent var	1.492507	
S.E. of regression	0.081986	Akaike info criterion	-2.078021	
Sum squared resid	0.201653	Schwarz criterion	-1.941975	
Log likelihood	37.28735	Hannan-Quinn criter.	-2.032246	
F-statistic	5287.374	Durbin-Watson stat	0.863887	
Prob(F-statistic)	0.000000			

四、方差膨胀因子

模型: $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$

OLS: $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i}$

方差膨胀因子越大，多重共线性越强。

$$Var(\hat{\beta}_1) = \frac{\sigma_u^2 / \sum \dot{x}_{1i}^2}{(1 - r_{12}^2)} = \frac{\sigma_u^2}{\sum \dot{x}_{1i}^2} \times VIF$$

称

$$VIF = \frac{1}{(1 - r_{12}^2)}$$

方差膨胀因子

若 **VIF = 3**，此时，相关系数的绝对值为多少？

K元回归对应估计量的方差膨胀因子:

模型: $y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + u_i$

OLS: $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \cdots + \hat{\beta}_k x_{ki}$

$$\Rightarrow \text{Var}(\hat{\beta}_j) = \frac{\sigma_u^2}{\sum_{i=1}^n \dot{x}_{ji}^2} \times \left(\frac{1}{1 - R_j^2} \right), \quad j = 1, 2, \dots, k$$

其中: $\hat{\beta}_j$ 的方差膨胀因子 $VIF_j = \frac{1}{1 - R_j^2}$, $j = 1, 2, \dots, k$

R_j^2 为第 j 个自变量对其余自变量进行回归的拟合优度

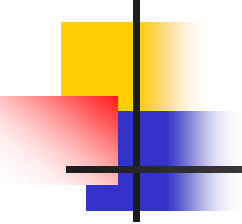
一般地, 若 **VIF** 大于**3** 以上, 表明存在严重的共线性。

(若是两个自变量, 此时, 相关系数的绝对值大于**0.82**



6.4 解决多重共线性的方法

- 1. 排除引起共线性的自变量。**
- 2. 差分法。**
- 3. 增加样本容量。**
- 4. 利用额外信息。**
- 5. 逐步回归法。**



1.排除引起共线性的自变量

找出引起多重共线性的自变量，将它排除出去，是克服多重共线性的最有效的方法。

但排除了引起共线性的变量后，保留变量的参数估计量可能是有偏的。

例6-1（续）

Dependent Variable: LOG(Y)
Method: Least Squares

排除自变量X2 的回归

Sample: 1980 2012
Included observations: 33

	Coefficient	Std. Error	t-Statistic	Prob.
C	4.186050	0.319516	13.10123	0.0000
LOG(X1)	0.563735	0.042974	13.11793	0.0000
LOG(X3)	0.215558	0.011784	18.29285	0.0000
R-squared	0.993725	Mean dependent var	11.83346	
Adjusted R-squared	0.993306	S.D. dependent var	0.539974	
S.E. of regression	0.044178	Akaike info criterion	-3.314653	
Sum squared resid	0.058552	Schwarz criterion	-3.178606	
Log likelihood	57.69177	Hannan-Quinn criter.	-3.268877	
F-statistic	2375.261	Durbin-Watson stat	0.218179	
Prob(F-statistic)	0.000000			

回归系数符合经济解释吗？回归系数显著吗？拟合效果如何？存在自相关吗？

Dependent Variable: LOG(Y)
Method: Least Squares

排除自变量X3 的回归

Sample: 1980 2012

Included observations: 33

	Coefficient	Std. Error	t-Statistic	Prob.
C	5.661477	0.436867	12.95928	0.0000
LOG(X1)	0.425435	0.056216	7.567909	0.0000
LOG(X2)	0.205597	0.012692	16.19887	0.0000
R-squared	0.992174	Mean dependent var	11.83346	
Adjusted R-squared	0.991653	S.D. dependent var	0.539974	
S.E. of regression	0.049334	Akaike info criterion	-3.093908	
Sum squared resid	0.073015	Schwarz criterion	-2.957862	
Log likelihood	54.04948	Hannan-Quinn criter.	-3.048133	
F-statistic	1901.802	Durbin-Watson stat	0.301690	
Prob(F-statistic)	0.000000			

回归系数符合经济解释吗？回归系数显著吗？拟合效果如何？存在自相关吗？

Dependent Variable: LOG(Y)
Method: Least Squares

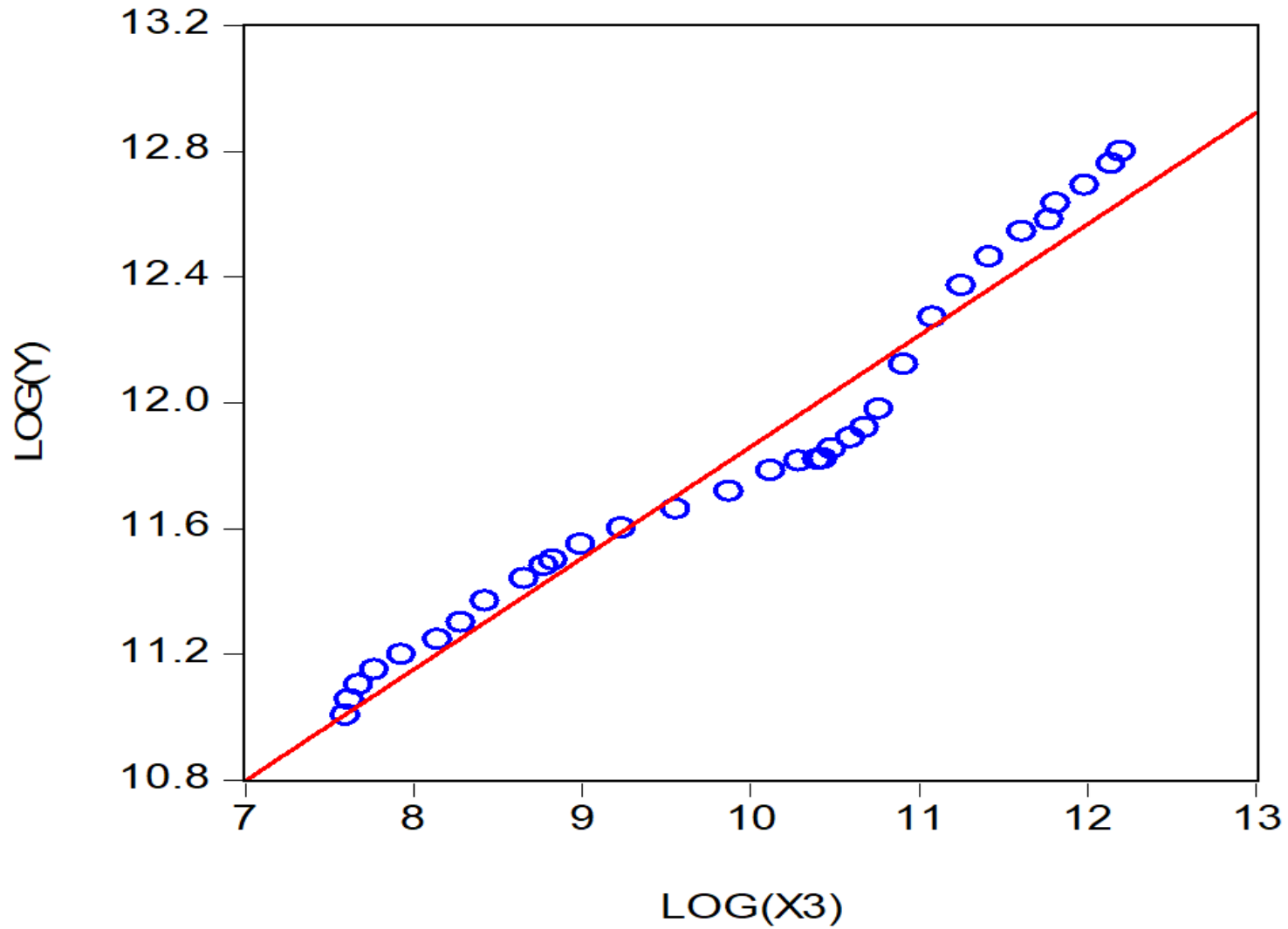
排除自变量X1 的回归

Sample: 1980 2012
Included observations: 33

	Coefficient	Std. Error	t-Statistic	Prob.
C	9.511808	0.203913	46.64630	0.0000
LOG(X2)	0.540165	0.083414	6.475753	0.0000
LOG(X3)	-0.299509	0.101306	-2.956481	0.0060
R-squared	0.982371	Mean dependent var		11.83346
Adjusted R-squared	0.981196	S.D. dependent var		0.539974
S.E. of regression	0.074046	Akaike info criterion		-2.281757
Sum squared resid	0.164484	Schwarz criterion		-2.145711
Log likelihood	40.64899	Hannan-Quinn criter.		-2.235981
F-statistic	835.8725	Durbin-Watson stat		0.438818
Prob(F-statistic)	0.000000			

回归系数符合经济解释吗？回归系数显著吗？拟合效果如何？存在自相关吗？

正相关还是负相关？



Correlation

	LOG(Y)	LOG(X1)	LOG(X2)	LOG(X3)
LOG(Y)	1.000000	0.961107	0.988552	0.978636
LOG(X1)	0.961107	1.000000	0.925367	0.896000
LOG(X2)	0.988552	0.925367	1.000000	0.996246
LOG(X3)	0.978636	0.896000	0.996246	1.000000

Dependent Variable: LOG(Y)

Method: Least Squares

Sample: 1980 2012

Included observations: 33

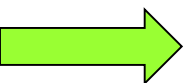
	Coefficient	Std. Error	t-Statistic	Prob.
C	8.320505	0.134001	62.09288	0.0000
LOG(X3)	0.354061	0.013360	26.50196	0.0000
R-squared	0.957728	Mean dependent var		11.83346
Adjusted R-squared	0.956365	S.D. dependent var		0.539974
S.E. of regression	0.112795	Akaike info criterion		-1.467792
Sum squared resid	0.394406	Schwarz criterion		-1.377095
Log likelihood	26.21857	Hannan-Quinn criter.		-1.437275
F-statistic	702.3540	Durbin-Watson stat		0.090696
Prob(F-statistic)	0.000000			

2. 差分法

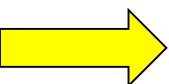
对于以时间序列数据为样本，利用差分法得到变量增量模型，一般自变量的增量之间的相关性较弱，能有效地避免出现多重共线性。

$$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \cdots + \beta_k x_{kt} + u_t$$

$$y_{t-1} = \beta_0 + \beta_1 x_{1t-1} + \beta_2 x_{2t-1} + \cdots + \beta_k x_{kt-1} + u_{t-1}$$

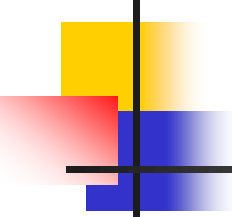


$$y_t - y_{t-1} = \beta_1 (x_{1t} - x_{1t-1}) + \beta_2 (x_{2t} - x_{2t-1}) + \cdots + \beta_k (x_{kt} - x_{kt-1}) + u_t - u_{t-1}$$



$$y_t^* = \beta_1 x_{1t}^* + \beta_2 x_{2t}^* + \cdots + \beta_k x_{kt}^* + u_t^*$$

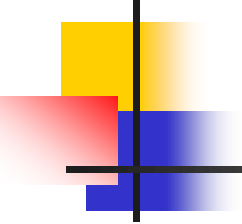
实际中如何设置此类差分模型更合理？



3.增加样本容量

由于多重共线性是样本引起的问题，增加一些不含共线特征的数据以扩大样本容量，可使参数估计量的方差减小，从而消除多重共线性造成的后果。

增加样本容量，对时间序列就是增大观测次数，对截面数据来说，就是增加观测对象，或者是把时间序列数据与截面数据结合起来使用。



4.利用额外信息

即利用样本数据之外的信息，如经济模型的理论假设关系、经济变量的经验关系与经验值等，将多元模型转化为一元模型，由此不必考虑多重共线性的问题。

在生产活动中，一般讲投入要素之间存在着较强的相关性，例如资本 K 与劳动 L 都与生产规模有关，二者之间也因此存在相关性。如以两要素的 C-D 生产函数为例，生产函数模型为

$$Y = AK^{\alpha}L^{\beta}e^u \quad (6-19)$$

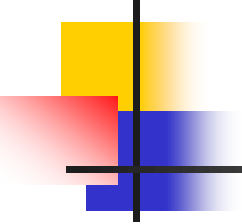
这里 Y 表示产出， K 为资本投入， L 为劳动投入。为避免 K 与 L 两解释变量出现共线性，可利用不变规模报酬的假设，即 $\alpha + \beta = 1$ ，将式 (6-19) 变换为

$$Y = AK^{\alpha}L^{1-\alpha}e^u$$

也即

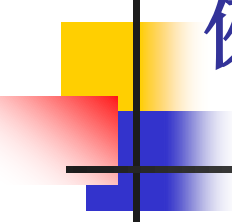
$$Y/L = A (K/L)^{\alpha} e^u \quad (6-20)$$

若以 K/L 为解释变量，以 Y/L 为被解释变量，式 (6-20) 即为一元模型，不存在共线性。



5.逐步回归法

在回归模型中,逐个引入自变量,根据拟合优度的变化决定是否引入新的自变量。如果增加一个自变量之后,模型的拟合优度有明显提高,并且各回归系数在统计上仍是显著的,则说明新引入的变量是一个独立的自变量;如果拟合优度变化不明显,或者显著地影响回归系数的数值或符号,或者致使原有自变量的方差(或标准差)显著增大、其显著性检验无法通过,这就表明新引入的自变量不能作为一个独立的自变量,它可能与其他自变量之间存在共线性关系。此时,只要把这种自变量排除出去,模型就避免了多重共线性。以此类推,从一元回归到多元回归,逐步得到有效的多元回归方程。此回归过程,称为逐步回归法。



例2: 逐步回归: 美国鸡肉需求模型

y —— 人均鸡肉消费量 (磅)

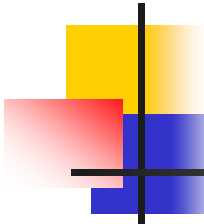
x_1 —— 人均实际可支配收入 (美元)

x_2 —— 鸡肉实际零售价格 (美分/磅)

x_3 —— 猪肉实际零售价格 (美分/磅)

x_4 —— 牛肉实际零售价格 (美分/磅)

数据来源: The data were collected by my student Robert J.Fisher and originally from the U.S.Department of Commerce



年	Y	X_1	X_2	X_3	X_4
1960	27.8	397.5	42.2	50.7	78.3
1961	29.9	413.3	38.1	52.0	79.2
1962	29.8	439.2	40.3	54.0	79.2
1963	30.8	459.7	39.5	55.3	79.2
1964	31.2	492.9	37.3	54.7	77.4
1965	33.3	528.6	38.1	63.7	80.2
1966	35.6	560.3	39.3	69.8	80.4
1967	36.4	624.6	37.8	65.9	83.9
1968	36.7	666.6	38.4	64.5	85.5
1969	38.4	717.8	40.1	70.0	93.7
1970	40.4	768.2	38.6	73.2	106.1
1971	40.3	843.3	39.8	67.8	104.8
1972	41.8	911.6	39.7	79.1	114.0
1973	40.4	931.1	52.1	95.4	124.1
1974	40.7	1021.5	48.9	94.2	127.6
1975	40.1	1165.9	58.3	123.5	142.9
1976	42.7	1349.6	57.9	129.9	143.6
1977	44.1	1449.4	56.5	117.6	139.2
1978	46.7	1575.5	63.7	130.9	165.5
1979	50.6	1759.1	61.6	129.8	203.3
1980	50.1	1994.2	58.9	128.0	219.6
1981	51.7	2258.1	66.4	141.0	221.6
1982	52.9	2478.7	70.4	168.2	232.6

Dependent Variable: Y
Method: Least Squares

一元模型回归

Sample: 1960 1982
Included observations: 23

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	27.97013	1.001010	27.94191	0.0000
X1	0.011303	0.000835	13.53348	0.0000
R-squared	0.897137	Mean dependent var	39.66957	
Adjusted R-squared	0.892239	S.D. dependent var	7.372950	
S.E. of regression	2.420319	Akaike info criterion	4.688617	
Sum squared resid	123.0169	Schwarz criterion	4.787356	
Log likelihood	-51.91910	F-statistic	183.1550	
Durbin-Watson stat	0.346259	Prob(F-statistic)	0.000000	

Dependent Variable: Y
Method: Least Squares

二元模型回归

Sample: 1960 1982
Included observations: 23

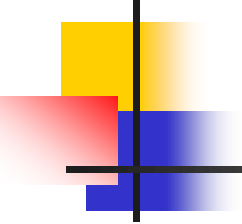
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	34.51460	3.855592	8.951829	0.0000
X1	0.014883	0.002193	6.785431	0.0000
X2	-0.213568	0.121900	-1.751997	0.0951
R-squared	0.910823	Mean dependent var	39.66957	
Adjusted R-squared	0.901906	S.D. dependent var	7.372950	
S.E. of regression	2.309209	Akaike info criterion	4.632795	
Sum squared resid	106.6489	Schwarz criterion	4.780903	
Log likelihood	-50.27714	F-statistic	102.1369	
Durbin-Watson stat	0.432817	Prob(F-statistic)	0.000000	

Dependent Variable: Y
Method: Least Squares

三元模型回归 (1)

Sample: 1960 1982
Included observations: 23

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	38.64639	3.649497	10.58951	0.0000
X1	0.010876	0.002381	4.567512	0.0002
X2	-0.541061	0.157968	-3.425136	0.0028
X3	0.174052	0.062530	2.783480	0.0118
R-squared	0.936654	Mean dependent var	39.66957	
Adjusted R-squared	0.926652	S.D. dependent var	7.372950	
S.E. of regression	1.996800	Akaike info criterion	4.377740	
Sum squared resid	75.75700	Schwarz criterion	4.575217	
Log likelihood	-46.34401	F-statistic	93.64706	
Durbin-Watson stat	0.882833	Prob(F-statistic)	0.000000	



随机项的自相关性检验

$$n = 23, \quad k' = 3, \quad \alpha = 0.05$$

$$d_L = 1.078, \quad d_U = 1.660$$

$$\therefore d = 0.8828 < d_L$$

$\therefore u_i$ 存在正自相关。

Dependent Variable: Y
Method: Least Squares

四元模型回归

Sample: 1960 1982

Included observations: 23

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	37.23220	3.717451	10.01552	0.0000
X1	0.005011	0.004892	1.024315	0.3193
X2	-0.611159	0.162848	-3.752932	0.0015
X3	0.198403	0.063719	3.113703	0.0060
X4	0.069498	0.050982	1.363178	0.1896
R-squared	0.942582	Mean dependent var	39.66957	
Adjusted R-squared	0.929822	S.D. dependent var	7.372950	
S.E. of regression	1.953174	Akaike info criterion	4.366448	
Sum squared resid	68.66797	Schwarz criterion	4.613295	
Log likelihood	-45.21416	F-statistic	73.87248	
Durbin-Watson stat	1.065025	Prob(F-statistic)	0.000000	

Dependent Variable: X1

Method: Least Squares

X1的共线性回归

Sample: 1960 1982

Included observations: 23

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-309.5749	159.1997	-1.944570	0.0668
X2	-8.568361	7.379145	-1.161159	0.2600
X3	6.809665	2.547070	2.673529	0.0150
X4	9.164150	1.138187	8.051535	0.0000
R-squared	0.981021	Mean dependent var	1035.074	
Adjusted R-squared	0.978024	S.D. dependent var	617.8416	
S.E. of regression	91.59057	Akaike info criterion	12.02930	
Sum squared resid	159387.8	Schwarz criterion	12.22678	
Log likelihood	-134.3370	F-statistic	327.3651	
Durbin-Watson stat	1.485835	Prob(F-statistic)	0.000000	

Dependent Variable: X2
Method: Least Squares

X2的共线性回归

Sample: 1960 1982
Included observations: 23

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	16.57324	3.601398	4.601890	0.0002
X1	-0.007733	0.006660	-1.161159	0.2600
X3	0.300068	0.057610	5.208613	0.0000
X4	0.098856	0.068148	1.450610	0.1632
R-squared	0.947094	Mean dependent var	47.99565	
Adjusted R-squared	0.938741	S.D. dependent var	11.11721	
S.E. of regression	2.751571	Akaike info criterion	5.018992	
Sum squared resid	143.8517	Schwarz criterion	5.216469	
Log likelihood	-53.71841	F-statistic	113.3769	
Durbin-Watson stat	1.385623	Prob(F-statistic)	0.000000	

Dependent Variable: X3
Method: Least Squares

X3的共线性回归

Sample: 1960 1982
Included observations: 23

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-17.30867	12.78174	-1.354172	0.1916
X1	0.040143	0.015015	2.673529	0.0150
X2	1.959946	0.376289	5.208613	0.0000
X4	-0.224311	0.176196	-1.273079	0.2183
R-squared	0.965577	Mean dependent var	90.40000	
Adjusted R-squared	0.960142	S.D. dependent var	35.22369	
S.E. of regression	7.032225	Akaike info criterion	6.895654	
Sum squared resid	939.5915	Schwarz criterion	7.093131	
Log likelihood	-75.30002	F-statistic	177.6530	
Durbin-Watson stat	1.197158	Prob(F-statistic)	0.000000	

Dependent Variable: X4
Method: Least Squares

X4的共线性回归

Sample: 1960 1982
Included observations: 23

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	20.34866	16.06361	1.266755	0.2206
X1	0.084388	0.010481	8.051535	0.0000
X2	1.008624	0.695311	1.450610	0.1632
X3	-0.350393	0.275233	-1.273079	0.2183
R-squared	0.974846	Mean dependent var	124.4304	
Adjusted R-squared	0.970874	S.D. dependent var	51.49974	
S.E. of regression	8.789107	Akaike info criterion	7.341674	
Sum squared resid	1467.720	Schwarz criterion	7.539151	
Log likelihood	-80.42925	F-statistic	245.4471	
Durbin-Watson stat	1.412595	Prob(F-statistic)	0.000000	



自变量相关系数矩阵

Correlation Matrix

	X1	X2	X3	X4
X1	1.000000	0.931676	0.957129	0.985875
X2	0.931676	1.000000	0.970112	0.928469
X3	0.957129	0.970112	1.000000	0.940567
X4	0.985875	0.928469	0.940567	1.000000

Dependent Variable: Y
Method: Least Squares

三元模型回归 (2)

Sample: 1960 1982
Included observations: 23

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	33.79811	4.286108	7.885501	0.0000
X1	0.012976	0.005035	2.577120	0.0185
X2	-0.222299	0.126181	-1.761745	0.0942
X4	0.024994	0.059084	0.423023	0.6770
R-squared	0.911655	Mean dependent var	39.66957	
Adjusted R-squared	0.897706	S.D. dependent var	7.372950	
S.E. of regression	2.358120	Akaike info criterion	4.710377	
Sum squared resid	105.6539	Schwarz criterion	4.907855	
Log likelihood	-50.16934	F-statistic	65.35562	
Durbin-Watson stat	0.395069	Prob(F-statistic)	0.000000	

Dependent Variable: Y
Method: Least Squares

三元模型回归 (3)

Sample: 1960 1982
Included observations: 23

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	27.10332	3.322019	8.158691	0.0000
X1	0.009737	0.006143	1.585070	0.1295
X3	0.015014	0.053141	0.282528	0.7806
X4	0.009081	0.062861	0.144468	0.8867
R-squared	0.897654	Mean dependent var	39.66957	
Adjusted R-squared	0.881494	S.D. dependent var	7.372950	
S.E. of regression	2.538117	Akaike info criterion	4.857493	
Sum squared resid	122.3987	Schwarz criterion	5.054970	
Log likelihood	-51.86117	F-statistic	55.54815	
Durbin-Watson stat	0.344369	Prob(F-statistic)	0.000000	

Dependent Variable: Y
Method: Least Squares

三元模型回归 (4)

Sample: 1960 1982
Included observations: 23

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	35.68084	3.399337	10.49641	0.0000
X2	-0.654097	0.157564	-4.151300	0.0005
X3	0.232528	0.054387	4.275460	0.0004
X4	0.115422	0.024303	4.749224	0.0001
R-squared	0.939235	Mean dependent var	39.66957	
Adjusted R-squared	0.929641	S.D. dependent var	7.372950	
S.E. of regression	1.955702	Akaike info criterion	4.336146	
Sum squared resid	72.67063	Schwarz criterion	4.533624	
Log likelihood	-45.86568	F-statistic	97.89329	
Durbin-Watson stat	1.251523	Prob(F-statistic)	0.000000	

此回归方程通过所有检验吗？是最好的回归结果吗

Breusch-Godfrey Serial Correlation LM Test:

F-statistic	1.427810	Probability	0.267182
Obs*R-squared	3.307842	Probability	0.191298

Test Equation:
Dependent Variable: RESID
Method: Least Squares

检验结果：不存在自相关吗

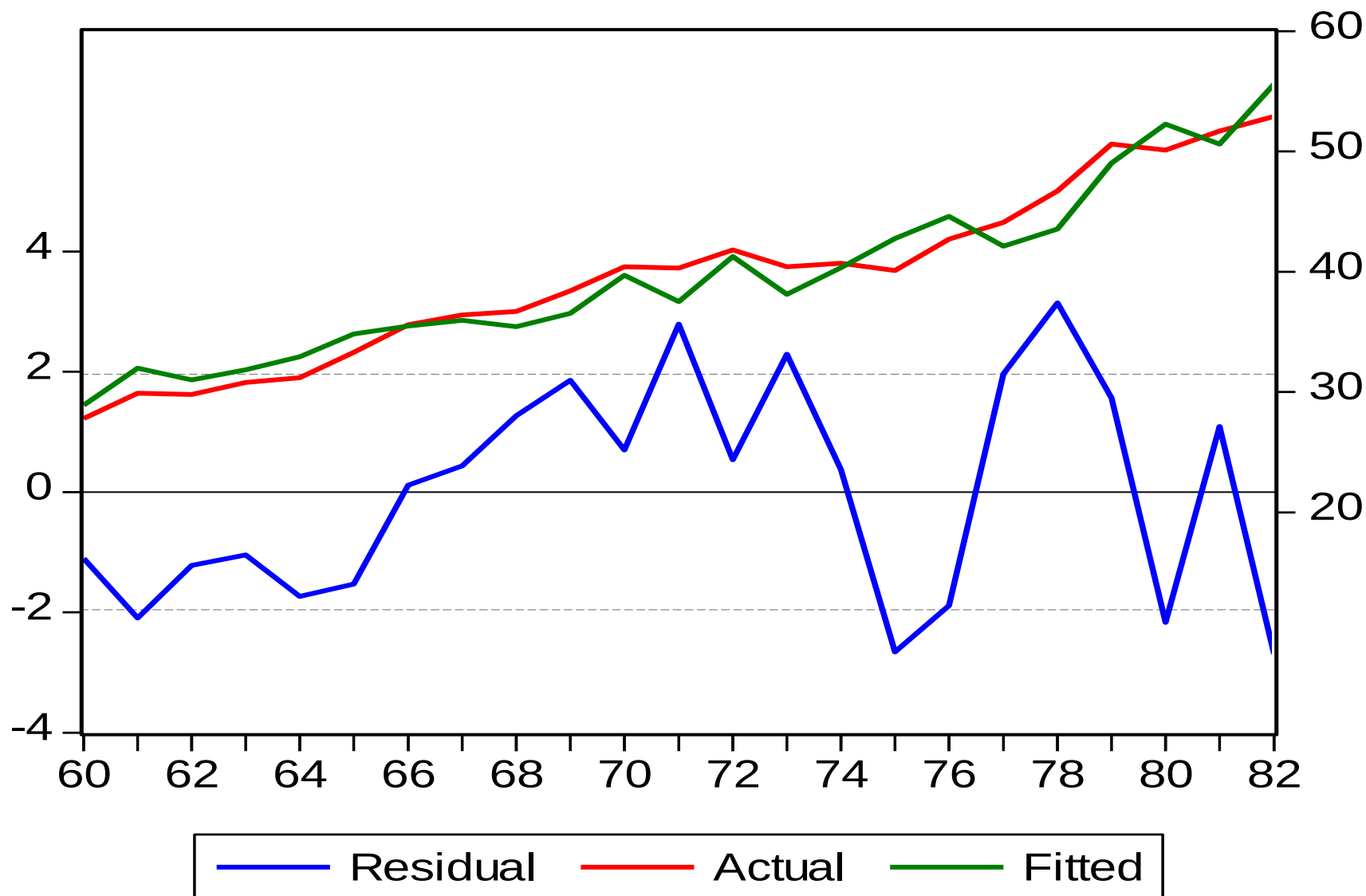
Presample missing value lagged residuals set to zero.

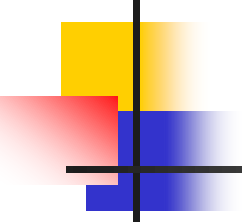
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-1.358629	3.421420	-0.397095	0.6962
X2	0.084036	0.162030	0.518644	0.6107
X3	-0.006215	0.054185	-0.114705	0.9100
X4	-0.017411	0.027536	-0.632318	0.5356
RESID(-1)	0.452304	0.278149	1.626124	0.1223
RESID(-2)	0.010288	0.280035	0.036738	0.9711

R-squared	0.143819	Mean dependent var	-4.60E-15
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随机项的自相关性检验

模型拟合效果





课外作业与上机作业 (教材**p113**)

课外作业: **1 , 2 , 3 , 4**

上机作业

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